

Master of Finance
Statistics & Econometrics
Solutions to Final Exam Dec 18, 2025

Problem 1: Test for Serial Correlation (2 credits)

Correct: The **Durbin–Watson** test can be used to detect serial correlation in regression residuals (especially AR(1)-type).

Why the others are wrong (quickly): Newey–West is a robust standard error correction (not a test statistic for serial correlation per se); Goldfeld–Quandt targets heteroskedasticity; Dickey–Fuller targets unit roots.

Problem 2: Intercept in a Linear Regression Model (2 credits)

Correct: In a linear regression model **without an intercept** the residuals are **not guaranteed** to be zero on average.

Reason: With an intercept, OLS enforces $\sum_t \hat{u}_t = 0$ (so $\bar{\hat{u}} = 0$). Without an intercept, this restriction does not hold in general.

Problem 3: Biases (2 credits)

Correct: Omitted variable bias can **mitigate** (attenuate) the estimated relationship between dependent and independent variable.

Reason: The sign of omitted-variable bias depends on the sign of (i) correlation between included regressor and omitted variable and (ii) the omitted variable's effect on y . Hence it can either inflate or mitigate the relationship.

Problem 4: Recession Prediction (8 credits)

Let $REC_{t+3} \in \{0, 1\}$ and $RetMkt_t$ denote annual log return.

(a) Model choice (2 credits)

Correct: A **Logit or Probit** model relating REC_{t+3} to $RetMkt_t$, both would give similar results.

(b) Interpreting coefficients (3 credits)

Given:

$$P(REC_{t+3} = 1 \mid RetMkt_t) = \Phi(\hat{\alpha} + \hat{\beta}RetMkt_t) = \Phi(-0.1 - 0.3 \cdot RetMkt_t).$$

Correct: The negative estimate $\hat{\beta}$ suggests that the recession probability **increases when market returns decline**.

(c) 50% recession calling rule (3 credits)

Call recession when model probability $> 50\%$. Since $\Phi(0) = 0.5$, we need:

$$-0.1 - 0.3 RetMkt_t > 0 \iff -0.3 RetMkt_t > 0.1 \iff RetMkt_t < -\frac{0.1}{0.3} = -0.333 \dots$$

Correct: $RetMkt_t$ would have to drop below negative 33 **percentage points**.

Problem 5: Confidence Interval (4 credits)

Regression output:

$$\hat{\beta} = 2.0, \quad SE(\hat{\beta}) = 0.8, \quad T = 2000.$$

A two-sided 90% CI uses $t_{crit} = 1.645$:

$$\hat{\beta} \pm 1.645 \cdot SE(\hat{\beta}) = 2.0 \pm 1.645 \cdot 0.8.$$

Lower bound:

$$2.0 - 1.645 \cdot 0.8 = 2.0 - 1.316 = 0.684 \approx 0.68.$$

Correct: 0.68.

Problem 6: F-Test (7 credits)

Unrestricted model:

$$r_{P,t} = \alpha + \beta_M MKT_t + \beta_{HML} HML_t + \beta_{SMB} SMB_t + u_t,$$

with $T = 120$, and $RSS_{UR} = \sum \hat{u}_t^2 = 130$.

(a) Restricted regression (3 credits)

Hypothesis: “comoves perfectly with HML” and “not exposed to market risk” means

$$H_0 : \quad \beta_M = 0, \quad \beta_{HML} = 1,$$

while leaving α and β_{SMB} unrestricted.

Move HML_t to the left:

$$r_{P,t} - HML_t = \alpha + \beta_{SMB} SMB_t + v_t.$$

Correct: A regression of $(r_P - HML)$ on SMB (with an intercept).

(b) F-statistic (3 credits)

Given restricted RSS $RSS_R = \sum \hat{v}_t^2 = 143$, number of restrictions $m = 2$, and unrestricted parameters $n = 4$ (intercept + 3 slopes), so $T - n = 116$.

$$F = \frac{(RRSS - URSS)}{URSS} \frac{(T - n)}{m} = \frac{(143 - 130)}{130} \frac{116}{2} = \frac{13 \cdot 58}{130} = 5.8$$

Correct: 5.8.

(c) Decision (1 credit)

Given critical values for $F_{2,116}$: 5% is 3.07 and 1% is 4.79. Since $5.8 > 4.79 > 3.07$, we reject at both levels.

Correct: The hypothesis can be rejected at both the 5% and the 1% level.

Problem 7: Hedge Portfolio (5 credits)

Regression (monthly excess returns):

$$r_{TMM,t} = \hat{\alpha} + \hat{\beta}r_{MKT,t} + \hat{u}_t = 0.9 + 0 \cdot r_{MKT,t} + \hat{u}_t.$$

(a) Average monthly TMM excess return (1 credit)

$$E[r_{TMM,t}] = \hat{\alpha} + \hat{\beta}E[r_{MKT,t}] = 0.9 + 0 \cdot E[r_{MKT,t}] = 0.9\%$$

Correct: $\hat{\alpha} = 0.9\%$.

(b) Alpha relative to NASDAQ (4 credits)

We want the intercept in the regression of r_{TMM} on r_{NAS} :

$$r_{TMM,t} = a + b r_{NAS,t} + \varepsilon_t.$$

Using summary stats,

$$b = \text{Corr}(r_{TMM}, r_{NAS}) \frac{\sigma_{TMM}}{\sigma_{NAS}} = 0.8 \cdot \frac{0.12}{0.08} = 0.8 \cdot 1.5 = 1.2.$$

With mean NASDAQ excess return $\bar{r}_{NAS} = 0.5\%$ and mean TMM excess return $\bar{r}_{TMM} = 0.9\%$,

$$a = \bar{r}_{TMM} - b \bar{r}_{NAS} = 0.9\% - 1.2 \cdot 0.5\% = 0.3\%$$

Correct: $\hat{\alpha} = 0.3\%$.

Problem 8: Stock Returns and Carbon Emissions (6 credits)

(a) Significance and sign (2 credits)

$$\hat{\gamma}_1 = 0.22, \quad SE(\hat{\gamma}_1) = 0.1 \quad \Rightarrow \quad t = \frac{0.22}{0.1} = 2.2.$$

Compare: 5% critical value ≈ 1.96 , 1% critical ≈ 2.326 . Thus significant at 5% but not at 1%.

Correct: Higher scope-1 emissions are associated with higher stock returns; statistically significant at 5% but not at 1%.

(b) Omitted variable bias logic (4 credits)

Multivariate regression:

$$r_{it} = \hat{\beta}_0 + \hat{\beta}_1 E_{it} + \hat{\beta}_2 S_{it} + \hat{v}_{it} = 3.5 + 0.06E_{it} + 2.1S_{it} + \hat{v}_{it},$$

with $SE(\hat{\beta}_1) = 1.2$, so

$$t_{\beta_1} = \frac{0.06}{1.2} = 0.05,$$

which is not statistically significant at conventional levels.

Since the univariate slope was 0.22 and the multivariate slope drops to 0.06, the univariate estimate was *biased upward*. A coherent explanation: sales S increases returns ($\hat{\beta}_2 > 0$) and is positively correlated with emissions E (larger firms emit more). Omitting S therefore pushes the coefficient on E upward.

Correct: The relationship between emissions and returns is biased upward in the first regression; in the second regression the emissions coefficient is not statistically different from zero.

Problem 9: Debt-to-GDP and Crisis Probability (6 credits)

Model:

$$P(CRISIS = 1 \mid DEBT) = \Phi(\alpha + \beta DEBT), \quad \hat{\alpha} = -1, \quad \hat{\beta} = 0.025.$$

(a) Probability at DEBT = 120 (3 credits)

Index:

$$z = -1 + 0.025 \cdot 120 = -1 + 3 = 2$$

Thus:

$$P(CRISIS = 1 \mid DEBT = 120) = L(2) = \frac{1}{1 + \exp(2)} \approx 0.881$$

Correct: $P(y_i = 1) = 88.1\%$.

(b) Marginal effect at DEBT = 80 (3 credits)

For a Logit model, the marginal effect is

$$\frac{\partial F}{\partial DEBT} = F(\alpha + \beta DEBT)(1 - F(\alpha + \beta DEBT)) \cdot \beta$$

At $\alpha + \beta DEBT = -1 + 0.025 \cdot 80 = 1$, this becomes

$$\frac{\partial F}{\partial DEBT} = 0.27 \cdot (1 - 0.27) \cdot 0.025 \approx 0.0049,$$

Correct: $ME = 0.49\%$

Problem 10: Panel Data Analysis

a. Two-way fixed effects specification

Answer: $y_{it} = \alpha_i + \mu_t + \beta x_{it} + u_{it}$.

b. Difference-in-differences estimate

$$\begin{aligned} \widehat{\text{DiD}} &= (\bar{Y}_{\text{After}}^{\text{Treated}} - \bar{Y}_{\text{Before}}^{\text{Treated}}) - (\bar{Y}_{\text{After}}^{\text{Control}} - \bar{Y}_{\text{Before}}^{\text{Control}}) \\ &= (7\% - 5\%) - (6\% - 5\%) = 2\% - 1\% = 1\%. \end{aligned}$$

Answer: $\widehat{\text{DiD}} = 1\%$.

Problem 11: Time Series

a. MA(1): γ_1 and ρ_1

Given $Y_t = \mu + u_t + \theta u_{t-1}$, $\theta = 0.2$, $\sigma_u^2 = 0.1$:

$$\gamma_1 = \text{Cov}(Y_t, Y_{t-1}) = \theta \sigma_u^2 = 0.2 \cdot 0.1 = 0.02.$$

$$\text{Var}(Y_t) = (1 + \theta^2) \sigma_u^2 = (1 + 0.04) \cdot 0.1 = 0.104.$$

$$\rho_1 = \frac{\gamma_1}{\text{Var}(Y_t)} = \frac{0.02}{0.104} \approx 0.192.$$

Answer: $\gamma_1 = 0.02$, $\rho_1 \approx 0.192$.

b. **Model from ACF/PACF pattern**

Geometrically decaying ACF with PACF cutoff after lag 1 is most consistent with an $AR(1)$.

Answer: $AR(1)$.

c. **Code blank for fitting $AR(p)$ via ARIMA class**

```
class statsmodels.tsa.arima.model.ARIMA(endog, exog=None, order=(0, 0, 0), ...)
because  $AR(p) \equiv ARIMA(p, 0, 0)$ .
```

Answer: $ARIMA(df, order=(p, 0, 0))$.

d. **Unconditional volatility of $AR(1)$**

For $Y_t = c + \phi Y_{t-1} + u_t$, $\phi = 0.95$, $\sigma_u^2 = 0.2$:

$$\text{Var}(Y_t) = \frac{\sigma_u^2}{1 - \phi^2} = \frac{0.2}{1 - 0.95^2} = \frac{0.2}{0.0975} \approx 2.051.$$

$$\text{Vol}(Y_t) = \sqrt{\text{Var}(Y_t)} \approx \sqrt{2.051} \approx 1.432.$$

Answer: $\text{Vol}(Y_t) \approx 1.432$.

e. **ARCH LM test**

$$LM = T \cdot R^2 = 240 \cdot 0.06 = 14.4.$$

Compare with $\chi_{0.95}^2(5) = 11.07$: $14.4 > 11.07$, so reject H_0 (no ARCH effects) at 5%.

Answer: $LM = 14.4$, do reject.

f. **AR(1)-ARCH(1): effect on unconditional variance**

With $\sigma_t^2 = \omega + \alpha u_{t-1}^2$, $\omega = 0.2$, $\alpha = 0.9$:

$$E[\sigma_t^2] = \frac{\omega}{1 - \alpha} = \frac{0.2}{0.1} = 2.0.$$

Answer: Increase.

g. **Rolling window code blank**

The blank corresponds to the first observation.

Answer: 0.

h. **Purpose of pseudo-out-of-sample exercise**

Answer: Overfitting and look-ahead bias.